Hall Ticket Number:

MR-21

Code: B315CS5S2

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. III Semester Supplementary Examinations, December 2025/January 2026

Discrete Mathematics

(Common to CSE, CSM and CSD)

Time: 3 Hours | ***Answer ALL Questions*** | Max. Marks: 70

Part-A (10 x 2 = 20 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	Define Tautology with a suitable example.	2	1	1	1
1. b)	Write contra positive of the conditional statement: p: 2+2=4, q: I am not the Prime Minister of India.	2	1	1	1
1. c)	Define Binary relation.	2	1	2	1
1. d)	If A={α,β}, B={1,2,3}. Find out (AxB) U (BxA) and (AxB) ∩ (BxA).	2	1	2	1,2
1. e)	What is recursive algorithm?	2	1	3	1
1. f)	Write the Principle of Mathematical Induction.	2	1	3	1
1. g)	Write the Fibonacci Recurrence Relation.	2	1	4	1
1. h)	What is Inclusion-Exclusion principle?	2	1	4	1
1. i)	What is Planar Graph?	2	1	5	1
1. j)	Define Shortest Path Algorithm.	2	1	5	1

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Prove that $(\forall x)(P(x)\vee Q(x))\Rightarrow(x)P(x)\vee(\exists x)Q(x)$	5	3	1	1,2
2. b)	Verify the following argument is valid by translating into symbols and using rules of inference If Clifton does not live in Franc, then he does not speak French Clifton does not drive a Datsum If Clifton lives in France, then he rides a bicycle Either Clifton speaks French, or he drives a Datsun Hence, Clifton rides a bicycle.	5	2	1	1,2
	OR				
2. c)	Show that $r\wedge(p\vee q)$ is a valid conclusion from the premises $(p\vee q),(q\rightarrow r),(p\rightarrow m)$ *and* $(\sim m)$.	5	2	1	1,2
2. d)	Show that (p→q) ∧ (q→r) → (p→r) is a Tautology.	5	2	1	1,2
3. a)	Let A = {1, 2, 3, 4} and R = {(1, 2), (2, 3), (3, 4), (2, 1)}. Determine the (i) Reflexive Closure of R (ii) Symmetric Closure of R (iii) transitive closure of R?	5	2	2	1,2
3. b)	Describe Hasse diagram? Let X = {2, 3, 6, 12, 24, 36}, and the relation ≤ be such that x ≤ y if x divides y. Draw the Hasse diagram of (X, ≤).	5	3	2	1,2
	OR				
3. c)	What are the truth sets of the Predicates of P(x),Q(x) and R(x), where the domain is the set of integers and P(x) is $\|x\|=1$,Q(x) is $x^2=2$ and R(x) is $\|x\|=x$.	5	2	2	1,2
3. d)	Find fog and gof, where $(x)=x^2+1$ and $g(x)=x+2$ are functions from R to R.	5	3	2	1,2
4. a)	Suppose the postal department prints only 5 and 9 cent stamps. Prove that it is possible to make up any postage of n cents using only 5 and 9 cent stamps for $n \geq 35$.	5	3	3	1,2
4. b)	Explain the principle of strong induction with example.	5	2	3	1

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	OR				
4. c)	Using induction principles prove that n^3+2n is divisible by 3.	5	4	3	1,2
4. d)	A class consists of 6 girls and 10 boys. If a committee of 3 is chosen at random from the class, find the probability that (i) 3 boys are selected (ii) exactly 2 girls are selected.	5	3	3	1,2
5. a)	Determine the solution of the recurrence relation $a_n = 6a_{n-1} - 9a_{n-2}$ with initial conditions $a_0 = 1$ and $a_1 = 6$?	5	4	4	1,2
5. b)	Let $X = \{1, 2, 3, 4, 5, 6, 7\}$ and $R = \{(x, y) \mid x-y \text{ is divisible by } 3 \text{ in } X\}$. Show that R is an equivalence relation.	5	3	4	1,2
	OR				
5. c)	Let R and S be the following relations $A= \{a,b,c,d\}$ defined $R= \{(a,a) ,(a,c) , (c,b) , (c,d),(d,b) \}$ and $S= \{ (b,a) , (c,c), (c,d) , (d,a)\}$ Find i) RoS ii) SoR iii) RoR.	5	3	4	1,2
5. d)	Solve the recurrence relation of the Fibonacci series of numbers.	5	3	4	1,2
6. a)	Show that a graph K_n has a Hamiltonian cycle whenever $n \geq 3$.	5	2	5	1,2
6. b)	A simple graph contains 35 edges, four vertices of degree 5, five vertices of degree 4 and four vertices of degree 3. Find the number of vertices with degree 2.	5	2	5	1
	OR				
6. c)	Show that K_5 is an Euler's circuit and also Hamiltonian cycle.	5	2	5	1
6. d)	Define the following with Diagram. (i) Chromatic Number (ii) Euler Path (iii) Euler circuit (iv) Hamiltonian cycle (v) coloring of a graph.	5	1	5	1

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome